

From Words to Worlds

How geospatial foundation models (AlphaEarth) are built — and how that mirrors LLMs

The one-sentence idea

Both an **LLM** and a **geospatial foundation model** do the same trick:

they take a firehose of raw, unlabeled data and learn to compress it into **embeddings** — vectors that place “similar things” near each other.

The mapping

LLM: *text* → token embeddings.

AlphaEarth: *a year of satellite/radar/lidar over a place* → a 64-number embedding.

The shared recipe

What a “foundation model” actually is

Three ingredients, identical across text and Earth observation:

1. **Massive unlabeled data.** The web for LLMs; petabytes of satellite imagery for AlphaEarth. Labels are scarce and expensive in both worlds.
2. **Self-supervised pre-training.** Invent a “fill-in-the-blank” task so the model teaches itself, with no human annotation.
3. **Reusable embeddings.** One pre-trained representation feeds *many* downstream tasks with little extra data.

The payoff (same in both fields)

Train once on everything; then solve new tasks with a handful of labels instead of thousands.

Primer: what is an embedding?

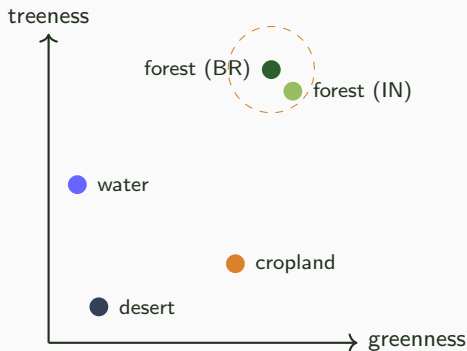
Imagine just two axes: **greenness** and **treeness**.

A satellite tile that is leafy and forested lands in the top-right. A desert tile lands near the origin.

Now scale up:

- not 2 axes but **64 dimensions** (“blueness”, “urbanness”, “smokeyness”...),
- not one tile but **billions** of them.

A forest in Indonesia and a forest in Brazil end up as *neighbors* in this space — even though the model was never told “forest”.



Similar places cluster together.

Two jobs an embedding does

1. Compress

Turn a huge stack of images (a whole year of Sentinel, Landsat, radar...) into a **small set of numbers**.

AlphaEarth: each 10 m pixel → **64 bytes**
— about $16\times$ smaller than comparable models.

2. Position

Place **similar** things close together, so the model captures *relationships*.

Exactly what word embeddings do:
“Albania” sits near “Albanian” and
“Angola” because they appear in similar contexts.

Same mathematical object; different raw material.

How an LLM is built (the reference)

How an LLM is trained — in four moves

1. **Tokenize.** Chop text into tokens (sub-words).
2. **Self-supervise.** Objective: predict the next (or a masked) token. The “label” is just the rest of the text — free, at web scale.
3. **Transformer + attention.** Each token’s representation is updated by *attending* to the others, so “bank” near “river” differs from “bank” near “money”.
4. **Adapt downstream.** Freeze the representation; fine-tune or few-shot it for translation, Q&A, classification...

Key: no human labels in step 2. The data supervises itself.

How AlphaEarth is built

AlphaEarth Foundations: a “virtual satellite”

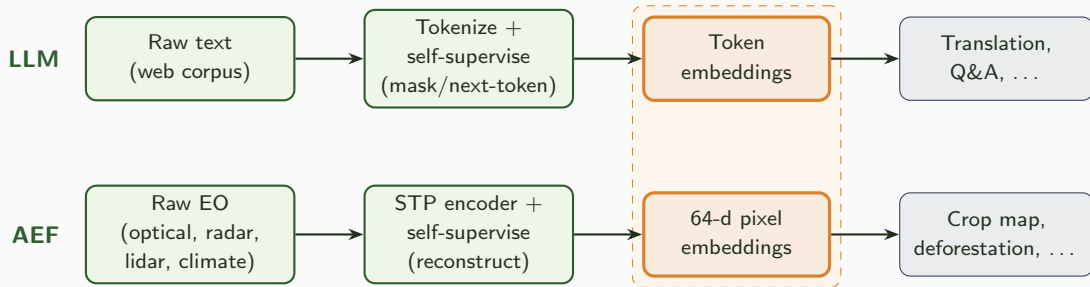
Google DeepMind (2025). What it produces:

- A global **embedding field**: every 10 m pixel gets a **64-dimensional** vector.
- Each vector is an **annual** summary — a temporal trajectory of surface conditions, not a single snapshot.
- Released free as an Earth Engine dataset (`GOOGLE/SATELLITE_EMBEDDING/V1`).

The problem it solves

Earth-observation data is vast but *labels are scarce*. Instead of building a bespoke model per task, learn *one* representation that fills gaps across space, time, and sensors.

The pipeline, side by side



The **embedding** stage is the shared heart of both systems.

The four tricks that make AlphaEarth work

1. **Continuous time.** Models time as a continuum, so it can *interpolate* gaps and *extrapolate* beyond observed dates — no need for complete sensor coverage.
2. **Space-Time Precision (STP) encoder.** Combines *spatial attention* (field shapes), *temporal attention* (how a place changes), and *convolutions* (fine detail), across scales.
3. **Contrastive learning with text.** Aligns imagery with geotagged descriptions (e.g. Wikipedia). A place tagged “soybean farm” links image $\leftrightarrow$ concept — this is the direct bridge to language models.
4. **Uniform embedding distribution.** Spreads vectors evenly on a unit hypersphere so they don’t “collapse”, improving generalization.

Attention: the piece both share

In an LLM

A word attends to other words in the sentence. Context disambiguates meaning: “*river bank*” vs. “*savings bank*”.

In AlphaEarth

A pixel attends to *neighboring pixels* (space) and to *the same pixel over time*: a burned patch is read in the context of what surrounded it and what it looked like last year.

The **transformer/attention** mechanism invented for language is the same engine that lets a geospatial model use spatial and temporal context.

The one big difference: the self-supervision signal

LLM

The free label is *the next token*. Language is a 1-D sequence; “predict what comes next” is natural.

AlphaEarth

There is no “next word”. The free labels come from *reconstruction* + *cross-sensor* + *contrastive-with-text* objectives: predict a held-out sensor, a held-out date, or match imagery to its geotagged caption.

Same *philosophy* (invent a task the data can grade itself on);
different *task*, because Earth is 2-D + time + many sensors, not a line of words.

Why it matters

Why the embedding approach wins

Benchmarked on 15 tasks (crop type, tree species, evapotranspiration, land-use change):

- **~24% lower error** than the next-best method at realistic dataset sizes.
- Strong even at **10-shot** and **1-shot** — exactly the label-scarce regime that matters for development economics.
- Predicts *continuous* biophysical variables (evapotranspiration $R^2 = 0.58$ where others failed).

For empirical research

An off-the-shelf 64-vector per pixel-year is a ready-made set of controls / features: land use, surface state, change — without hand-crafting NDVI-style indices.

What you can do with the embeddings

Similarity search

Pick a patch (farmland, forest); find all places on Earth with a similar surface signature.

Change detection

Difference a pixel's embedding across years → urban growth, wildfire, reservoir levels.

Clustering (no labels)

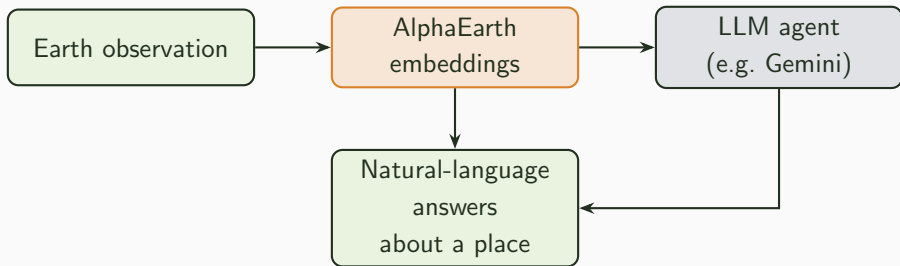
Group similar pixels → forest types, soil zones, urban styles.

Few-shot mapping

Train a map with *hundreds*, not tens of thousands, of labels.

These are the geospatial analogues of “prompt the LLM” — reuse without retraining.

The loop is closing: embeddings + LLM agents



DeepMind's stated direction: combine the annual embeddings with **general-reasoning LLM agents**. The geospatial model becomes the “eyes”, the LLM the “reasoner” — two foundation models speaking through a shared vector language.

Takeaways

1. **Same recipe:** massive unlabeled data → self-supervised pre-training → reusable embeddings.
2. **Same engine:** the transformer/attention mechanism from language models powers spatial + temporal context in AlphaEarth.
3. **Different signal:** “next token” for text; reconstruction + cross-sensor + contrastive-with-text for Earth.
4. **Same payoff:** solve new tasks with few labels — decisive where ground truth is scarce.
5. **Practical:** 64-d annual embeddings are free in Earth Engine; treat them as ready-made features for maps, change detection, and empirical work.

Questions?

References & further reading i

- Brown, Kazmierski, Pasquarella et al. (2025). *AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data*. arXiv:2507.22291.
- Google DeepMind (2025). *AlphaEarth Foundations helps map our planet in unprecedented detail* (blog).
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GOOGLE/SATELLITE_EMBEDDING/V1/ANNUAL.
- Iddawela, Y. (2025). *Google DeepMind's new geospatial model explained intuitively*. The Spatial Edge.
- Background on attention: Vaswani et al. (2017). *Attention Is All You Need*.